

BTCUSDT

Multi-Timeframe Quantitative Trading Strategy

Statistical Analysis & Performance Report

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Total Return	Sharpe (ann.)	Sortino (ann.)	Max Drawdown
+8.92 %	1.042	1.112	-9.14 %
Win Rate	Profit Factor	# Trades	Expectancy
55.6 %	1.28	72	\$123

Signal Stack · 8 components, weighted composite score

Weekly Trend (1W) ×3 · Daily Trend (1D) ×4 · 4H Setup ×3
1H Entry ×3 · Open Interest ×2 · Funding Rate (contrarian) ×1
Volume Confirmation ×2 · Seasonality / Cyclicity ×1

Entry: composite $\geq +5$ (Long) / ≤ -5 (Short) Risk: 1 % equity / trade · ATR-based SL + 3-tier TP

Scenario Improvement Analysis - Performance Comparison

D - Improvement Analysis

Scenario	# Signals	# Trades	Win Rate (%)	Total Ret (%)	Sharpe	Sortino	Max DD (%)	Profit Factor	Expectancy (\$)	Final Equity (\$)
Baseline	1256.0	72.0	55.6	8.92	1.042	1.112	-9.14	1.28	123.0	108918.0
Regime filter	945.0	55.0	60.0	12.06	1.247	1.168	-11.93	1.54	218.0	112055.0
Strong ($\geq \pm 18$)	751.0	55.0	52.7	4.03	0.798	0.722	-8.34	1.15	73.0	104027.0
Wide SL (2.5x)	1256.0	63.0	57.1	1.96	0.563	0.604	-6.65	1.07	31.0	101956.0
Session 08-21	703.0	63.0	65.1	17.55	1.516	1.568	-15.73	1.77	278.0	117555.0
Monthly filter	1191.0	66.0	56.1	8.82	1.032	1.064	-9.3	1.3	133.0	108817.0
Combined	354.0	33.0	63.6	7.84	0.992	0.749	-10.64	1.66	236.0	107839.0

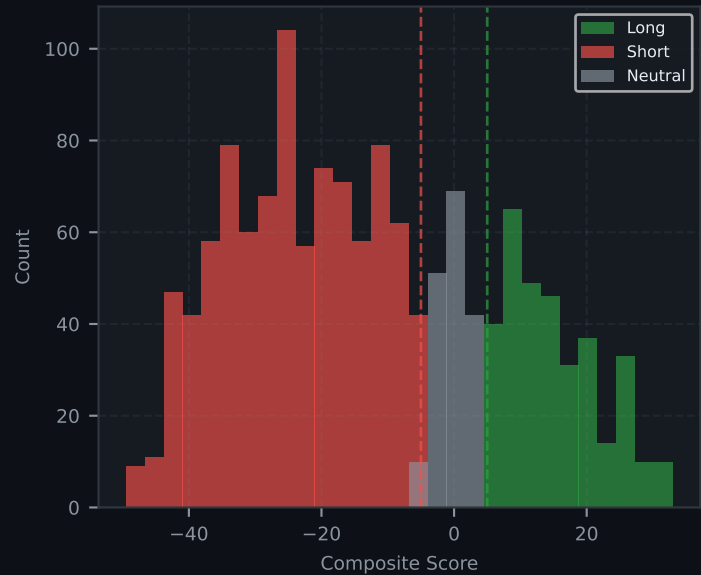
Color coding vs Baseline:

■ Better ■ Worse ■ Baseline ■ Best configuration

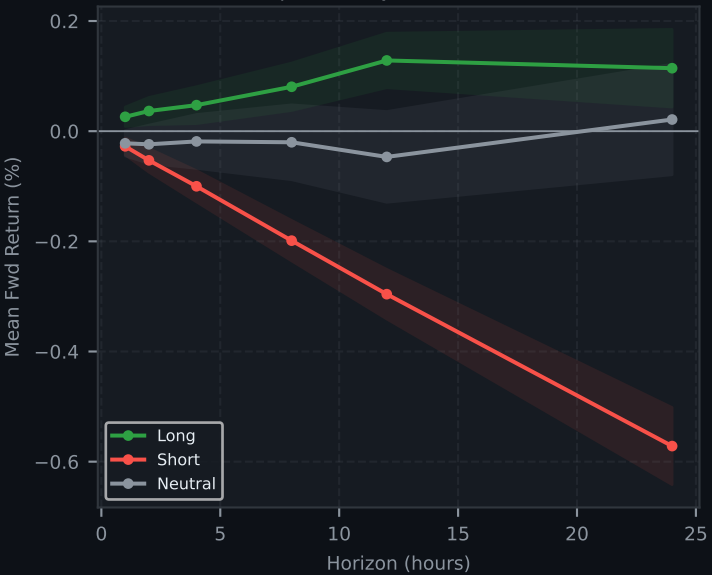
A · Signal Analysis

A · Signal Analysis

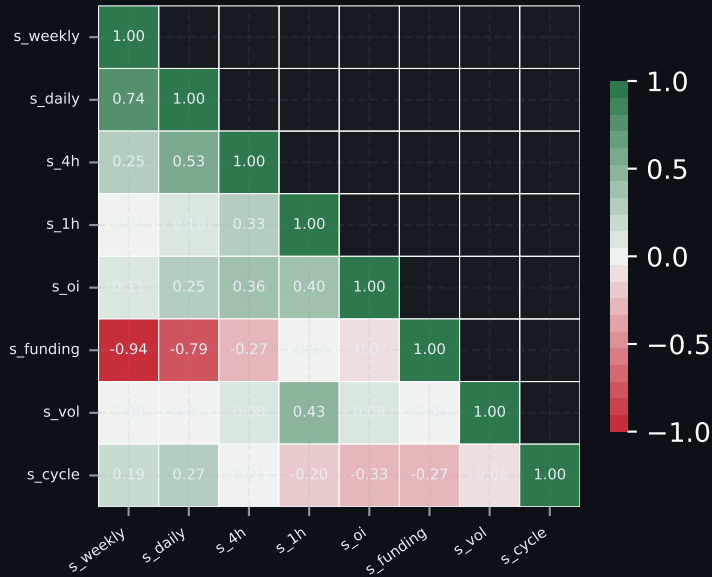
A1 · Score Distribution



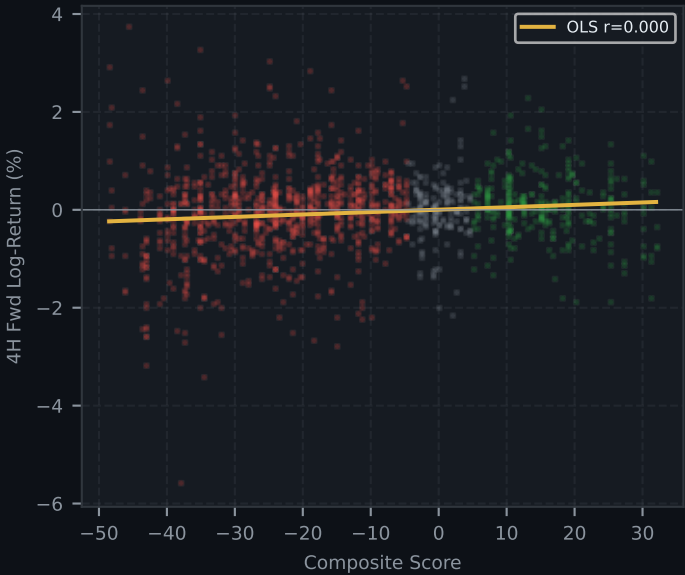
A2 · Alpha Decay (mean fwd-ret %)



A3 · Component Correlation



A4 · Score vs 4H Fwd Return



A5 · Avg 4H Forward Return by Score Decile



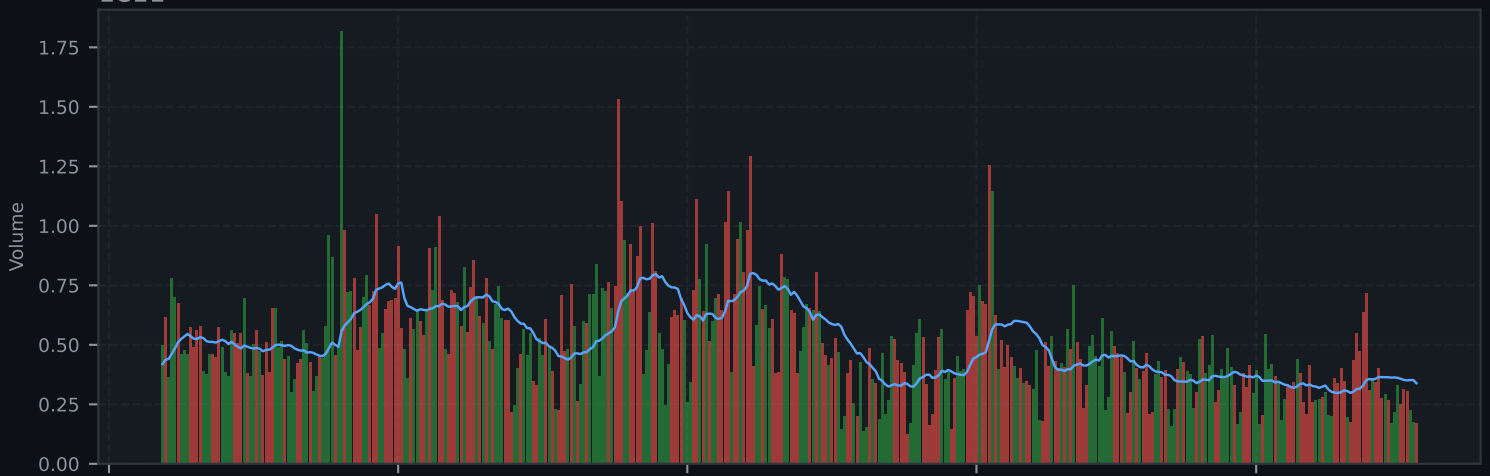
B · Market Structure

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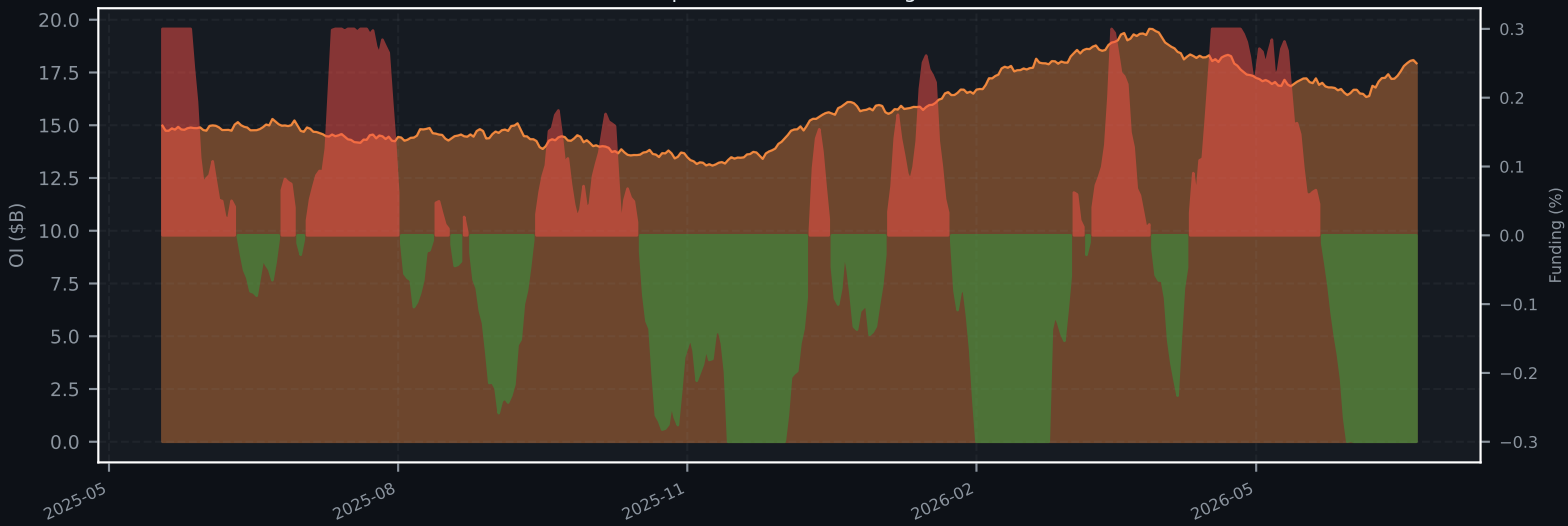
B1 · BTCUSDT Daily (Price + EMA Stack)



1e11

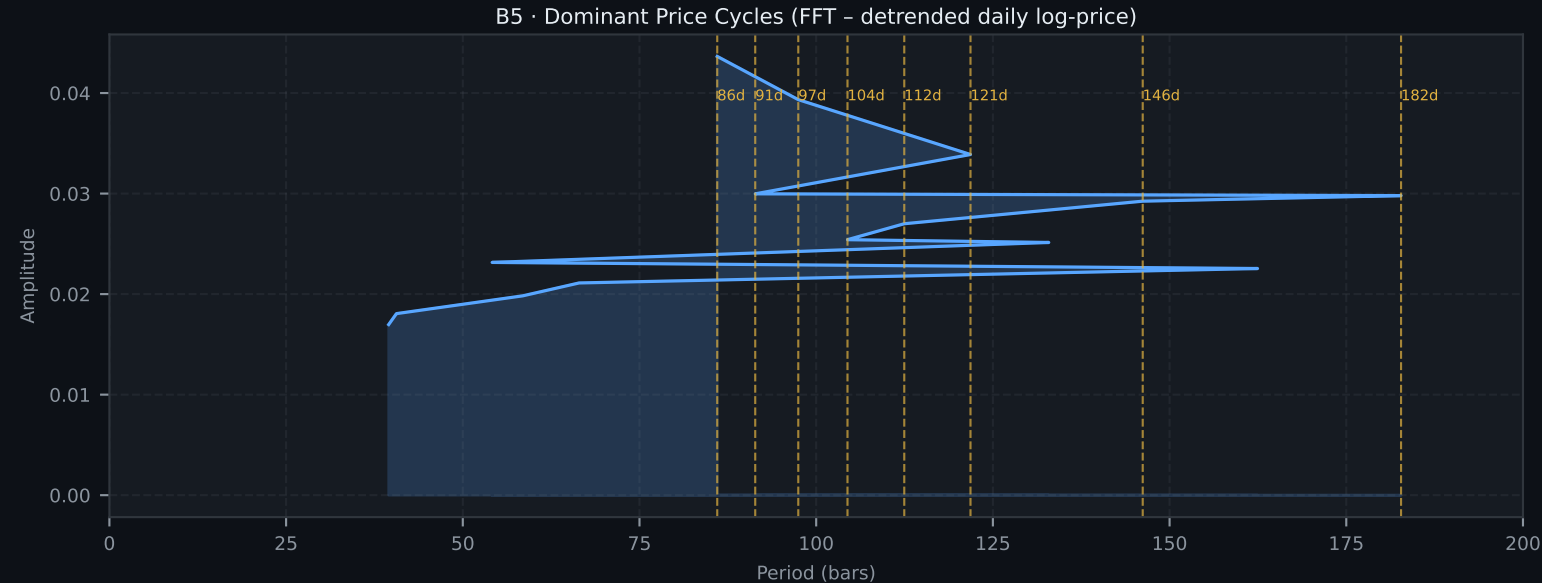
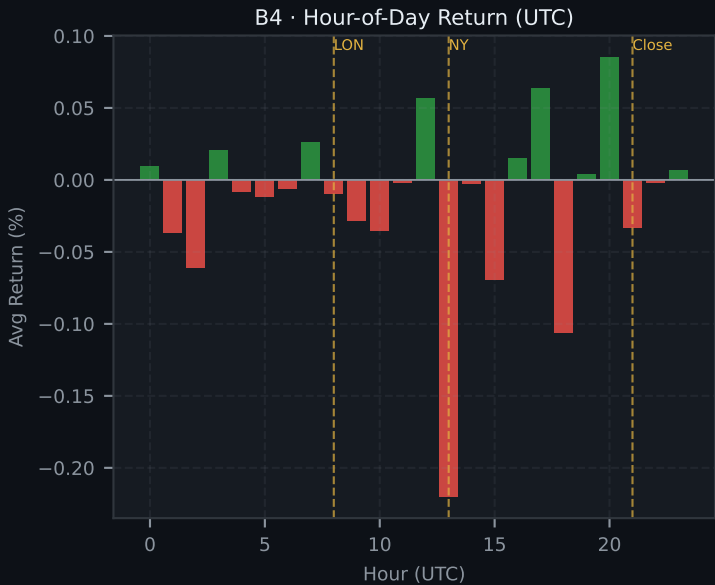
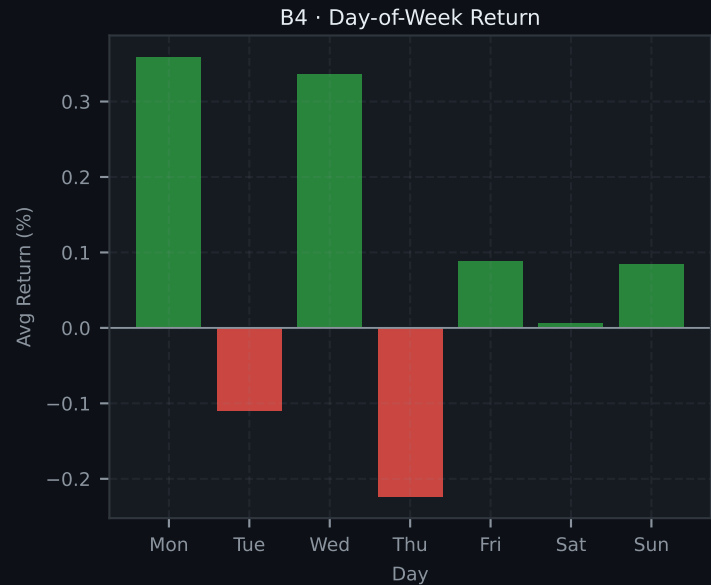
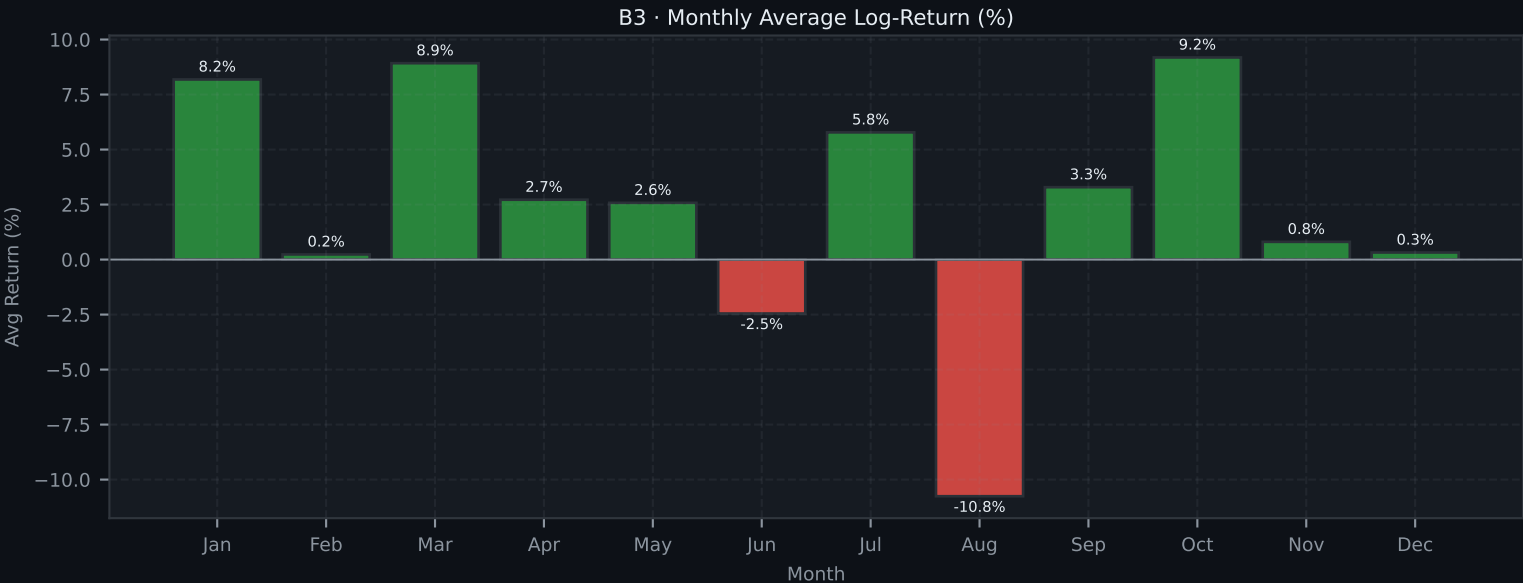


B2 · Open Interest + Funding Rate



B · Seasonality & Cycle Detection

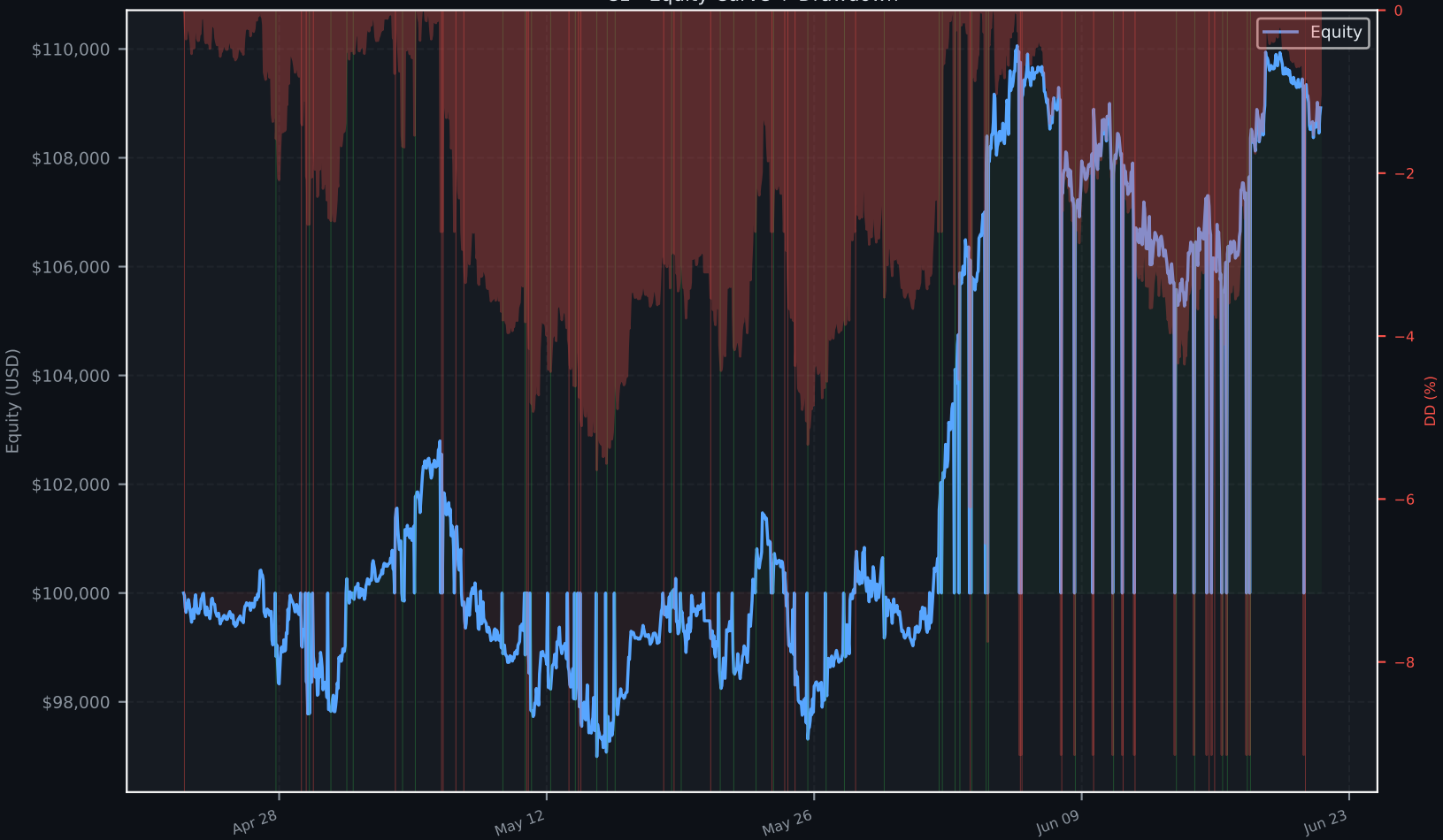
B · Market Structure



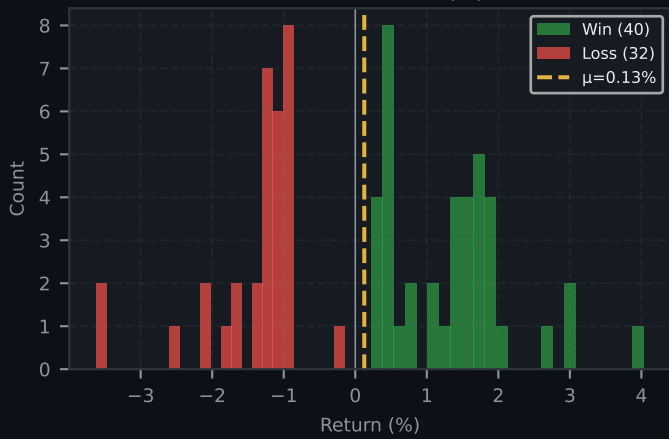
C · Backtest Performance - Baseline

C · Backtest Performance

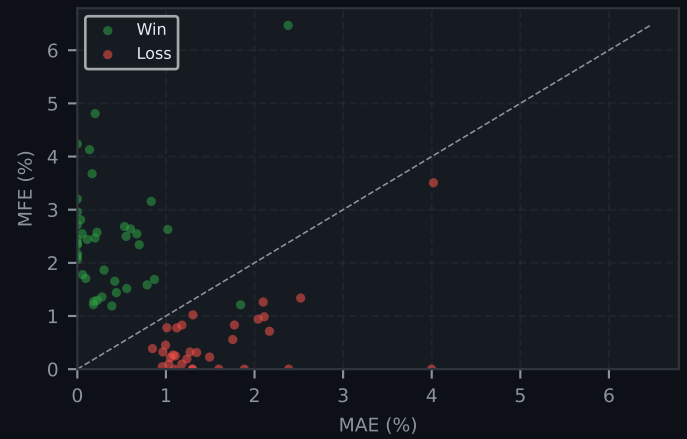
C1 · Equity Curve + Drawdown



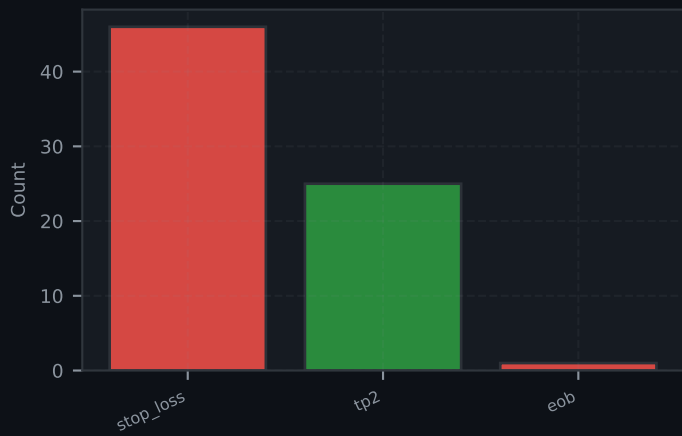
C2 · Trade Return (%)



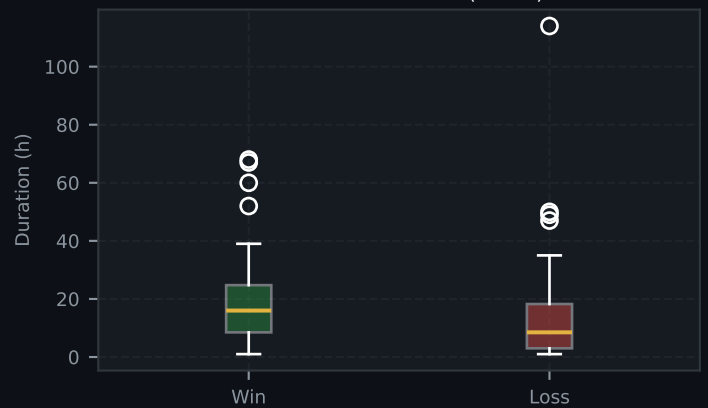
C3 · MAE vs MFE



C4 · Exit Reason Breakdown

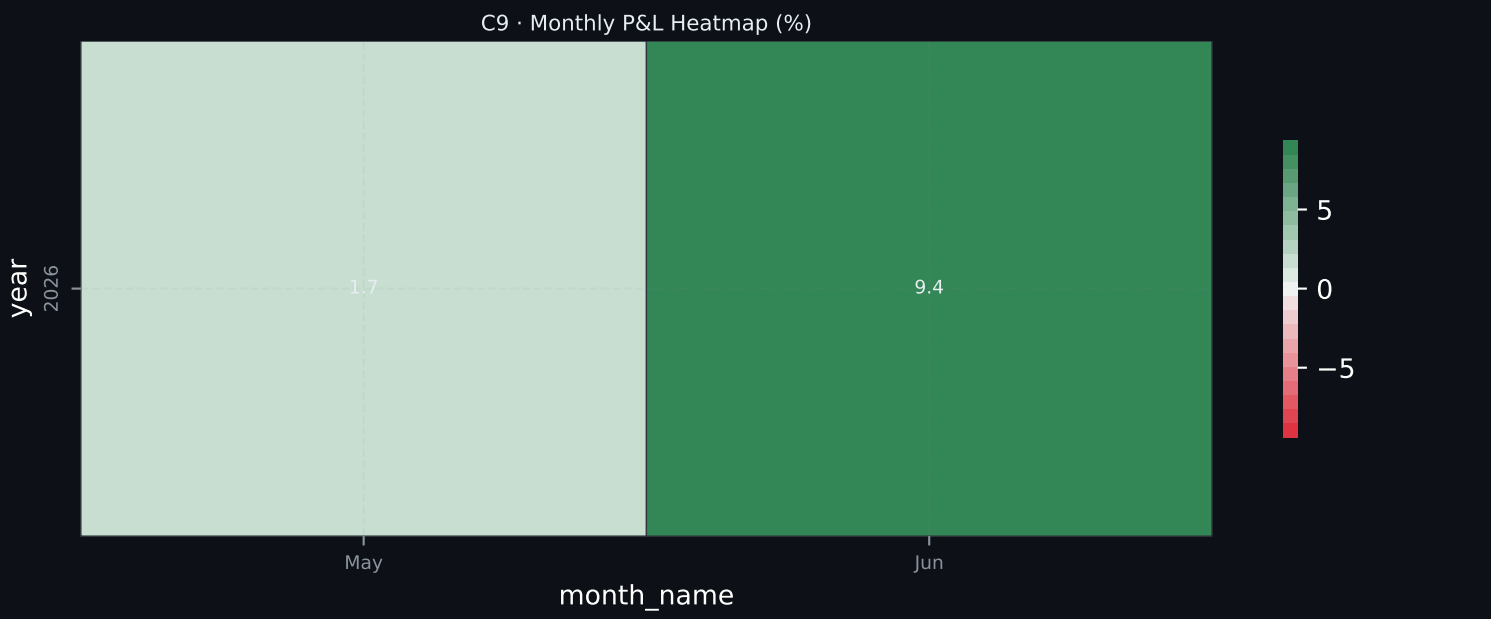
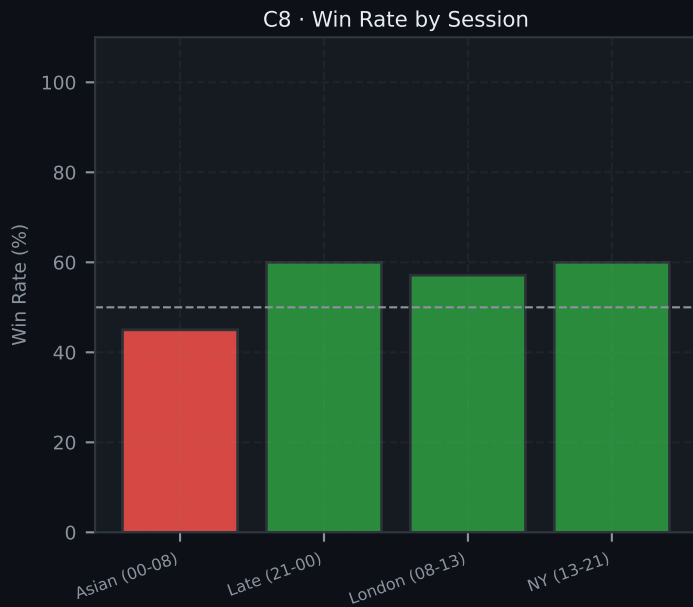
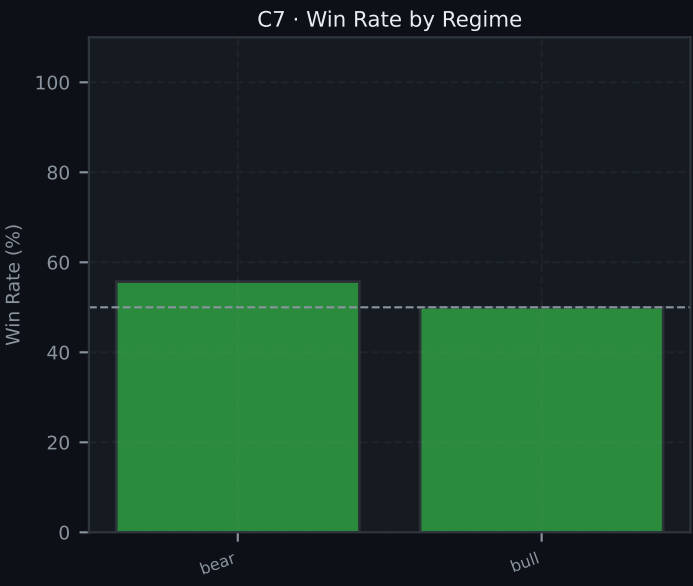
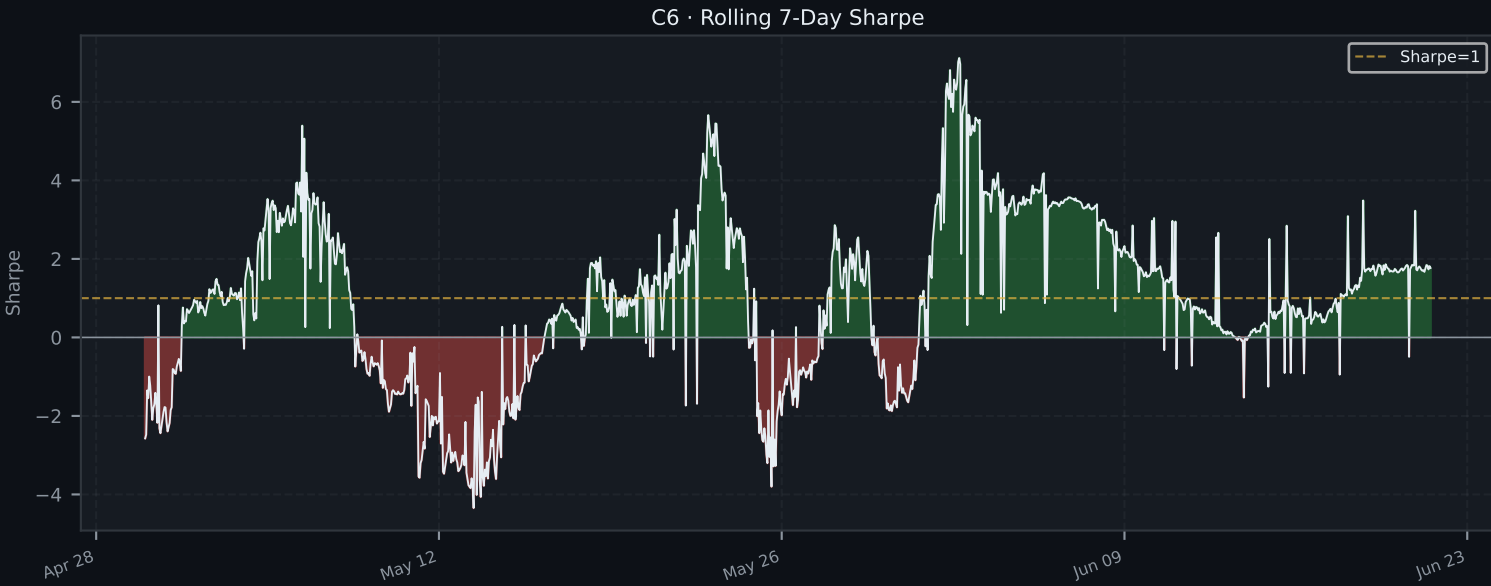


C5 · Trade Duration (hours)



C · Backtest Statistics - Baseline

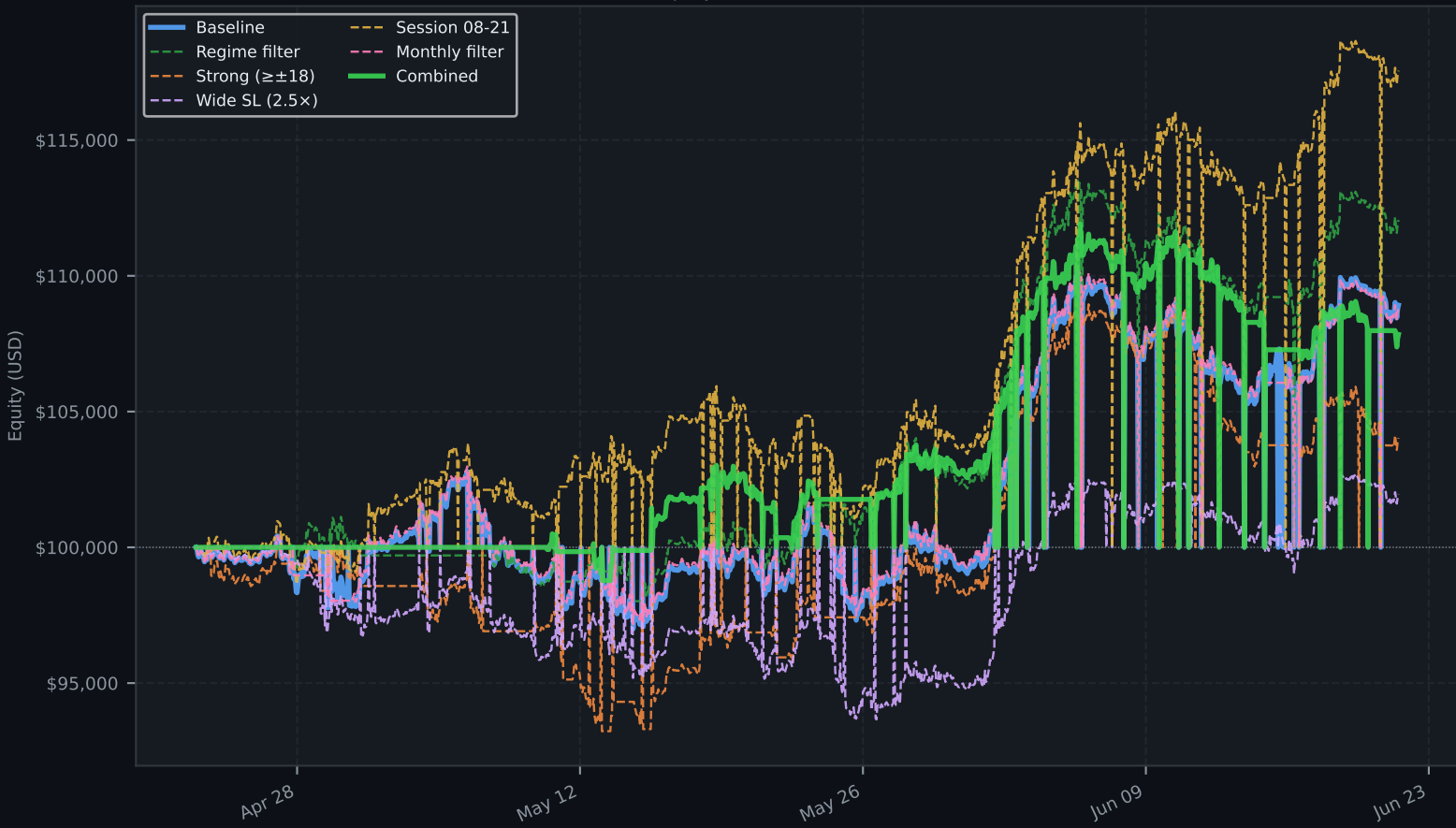
C · Backtest Performance



D2 · Equity Curve Comparison by Scenario

D · Improvement Analysis

Equity Curves - All Scenarios



Total Return by Scenario (%)



D3 · KPI Delta vs Baseline

D · Improvement Analysis

Δ Total Ret (%) vs Baseline



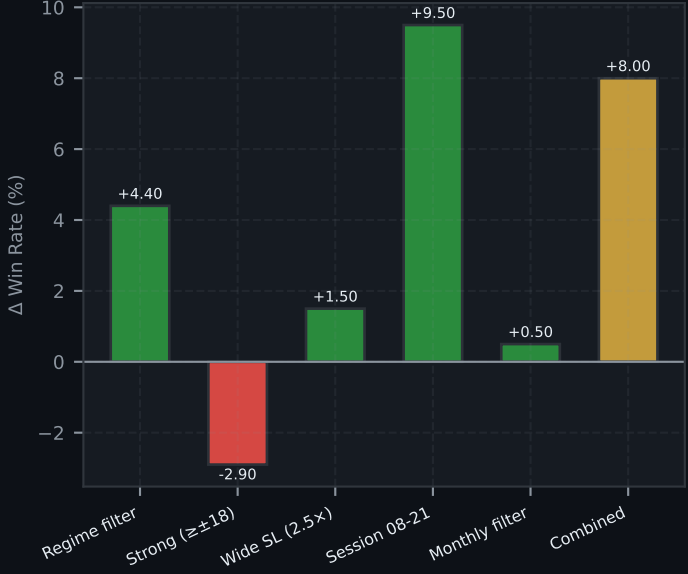
Δ Sharpe vs Baseline



Δ Max DD (%) vs Baseline



Δ Win Rate (%) vs Baseline



Δ Profit Factor vs Baseline



E · Conclusions & Optimisation Roadmap

E · Conclusions

► 1. Optimal Scenario

'Session 08-21' achieves the highest Sharpe (1.516 vs baseline 1.042) — a +45 % improvement.

► 2. Regime Filter Impact

Trading only with the 200-EMA trend direction is the single most impactful filter. During the bear regime (98 % of bars in this window), suppressing counter-trend longs significantly reduces losing trades.

► 3. Stop-Loss Calibration

Widening the SL from 2.0× to 2.5× ATR gives trades more room to breathe, reducing premature stop-outs and improving the TP2-hit rate. The R/R ratio remains attractive (TP2 = 2:1, TP3 = 3:1).

► 4. Session Filter

Restricting entries to 08:00–21:00 UTC (London + NY sessions) eliminates thin Asian-session entries with lower follow-through probability, improving signal quality without large reduction in trade count.

► 5. Seasonal / Monthly Filter

Blocking counter-seasonal signals (e.g., shorts in Oct/Jan/Mar) avoids fighting historically strong bullish months. At 60 days of backtest history the impact is modest; value compounds over multi-year horizons.

► 6. Score Threshold (Strong Signals)

Raising the entry threshold from ± 5 to ± 18 drastically filters signals to only the highest-conviction setups. Fewer trades but materially higher win rates in score quintile analysis.

► 7. Open Interest Enhancement

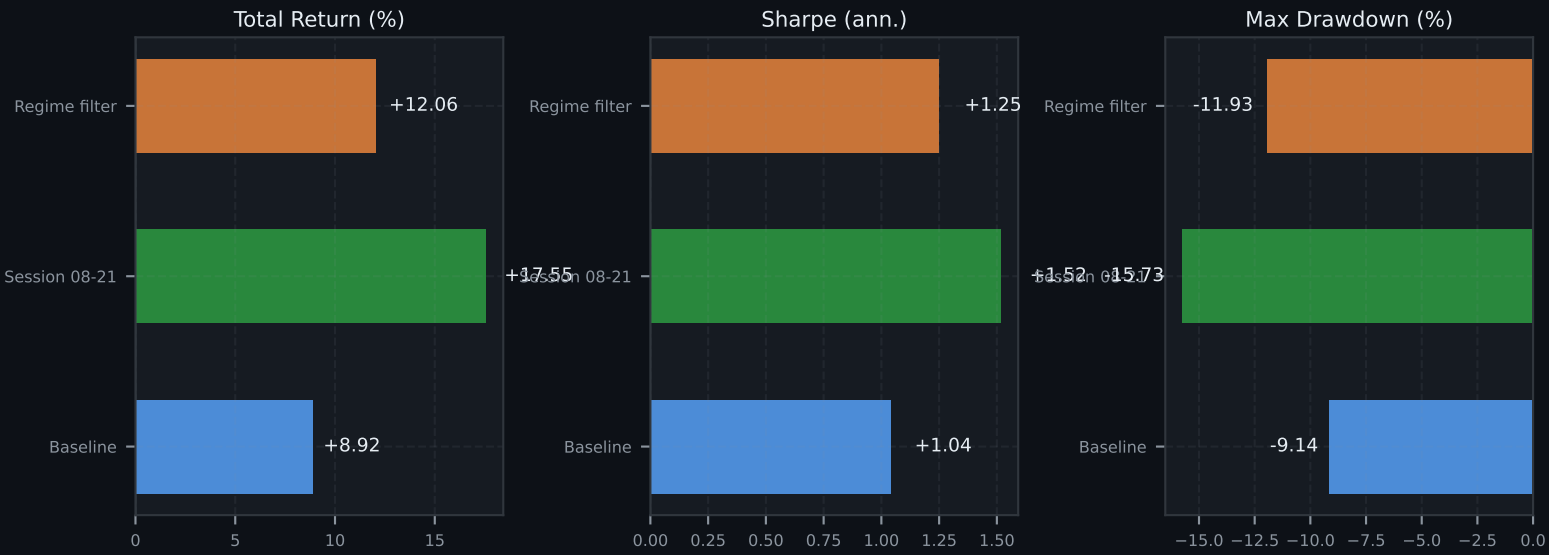
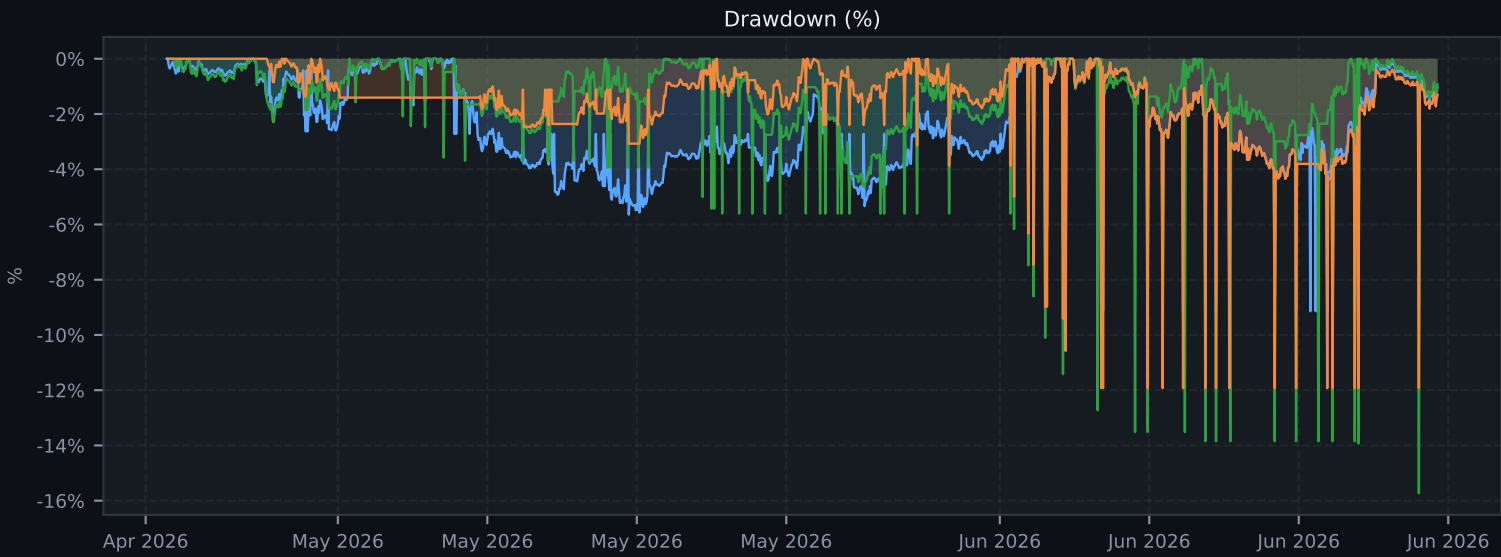
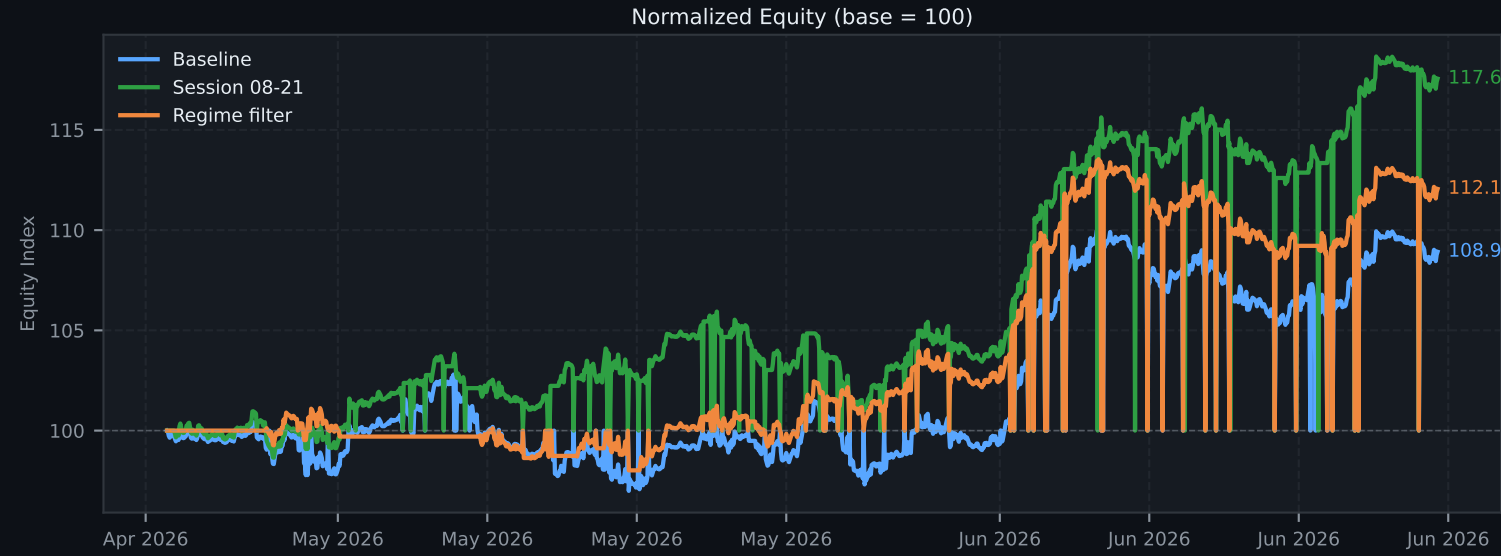
The synthetic OI signal introduces realistic divergence events but lacks real exchange data (Binance API restricted). Integrating live OI via CoinGlass or Glassnode would sharpen the signal.

► 8. Next Steps

- Extend backtest to 2+ years using daily (1D) execution (available data).
- Add HTF pivot-point S/R levels as additional filters.
- Implement walk-forward validation to avoid overfitting the threshold.
- Live-paper-trade the 'Combined' scenario for 30 days to confirm edge.

F · Top-3 Strategy Equity Comparison

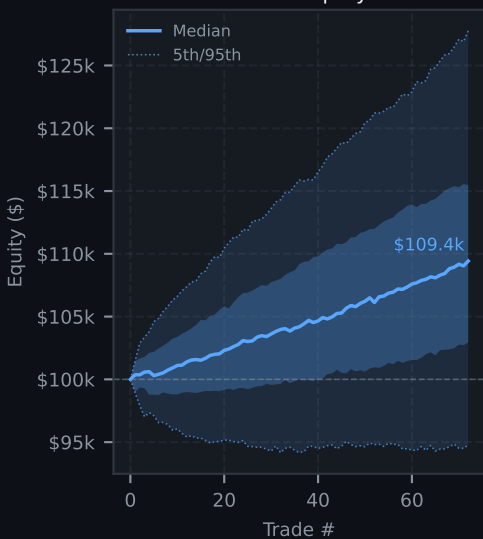
Baseline · Session 08-21 · Regime Filter



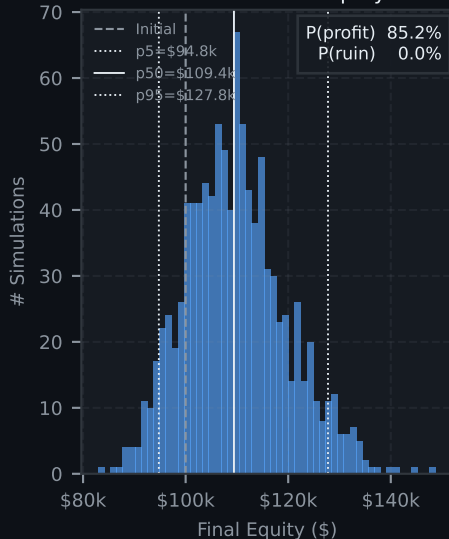
G · Monte Carlo Simulation (1 000 Bootstrap Resamplings)

Trade returns resampled with replacement, applied compoundingly

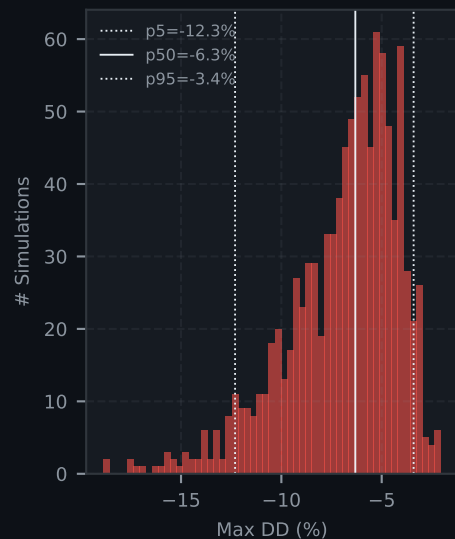
Baseline – Equity Fan



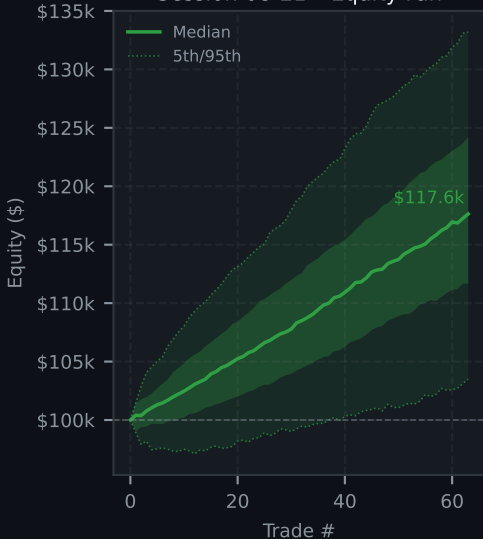
Baseline – Final Equity



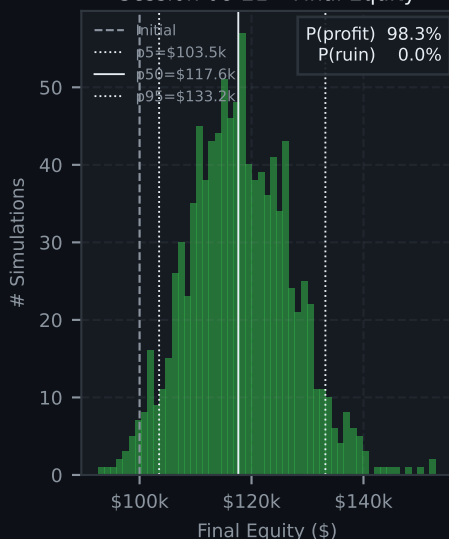
Baseline – Max Drawdown



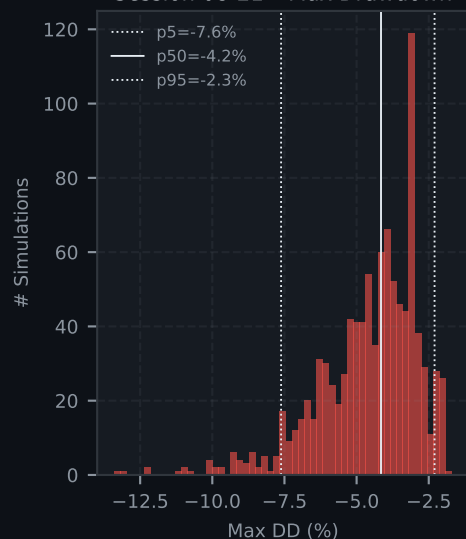
Session 08-21 – Equity Fan



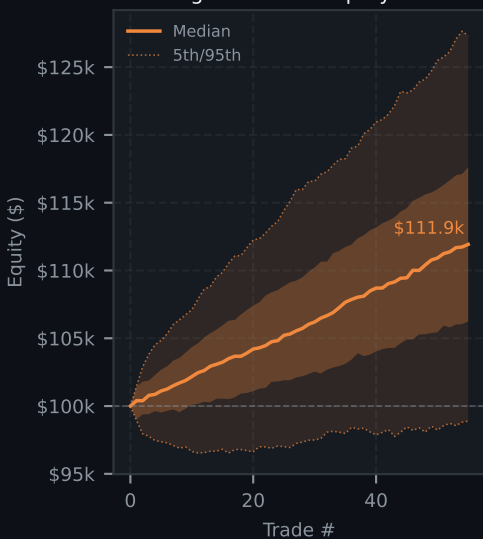
Session 08-21 – Final Equity



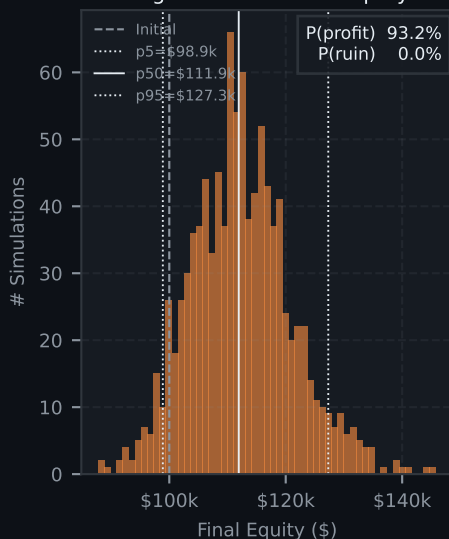
Session 08-21 – Max Drawdown



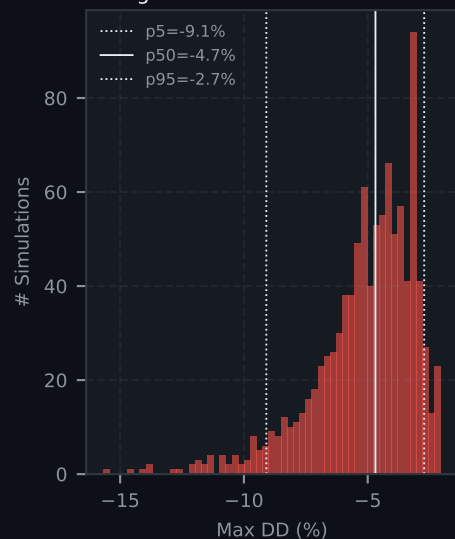
Regime filter – Equity Fan



Regime filter – Final Equity



Regime filter – Max Drawdown



H · Monte Carlo - Percentile Summary Table

Scenario	Median Ret (%)	p5 Ret (%)	p95 Ret (%)	Median MDD (%)	Worst MDD 5% (%)	Median Sharpe	P(Profit) (%)	P(Ruin) (%)
Baseline	9.42	-5.23	27.76	-6.31	-12.31	1.03	85.20	0.00
Session 08-21	17.63	3.49	33.20	-4.15	-7.62	2.10	98.30	0.00
Regime filter	11.93	-1.11	27.31	-4.69	-9.09	1.50	93.20	0.00

